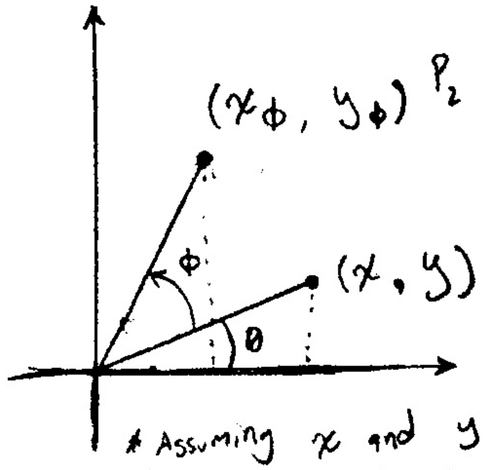
origin rotation



* Assuming x and y are known/defined

$$x = \cos(\theta)$$
$$x_\phi = \cos(\phi + \theta)$$
$$y = \sin(\theta)$$
$$y_\phi = \sin(\phi + \theta)$$

Sine / cosin Addition

$$\sin(a+b) = \sin a \cdot \cos b + \cos a \cdot \sin b$$
$$\cos(a+b) = \cos a \cdot \cos b - \sin a \cdot \sin b$$

Derived

Rotation rule

$$x' = x\cos(\theta) - y\sin(\theta)$$
$$y' = x\sin(\theta) + y\cos(\theta)$$

* If x and y are known you can rotate a point or most functions by θ using this equation

$$x_\phi = \cos(\phi)\underbrace{\cos(\theta)}_{x} - \sin(\phi)\underbrace{\sin(\theta)}_{y}$$

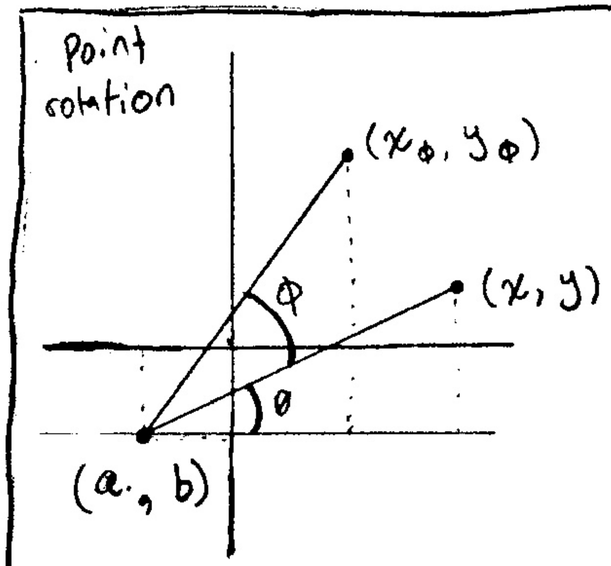
* This is just rotation around the origin

$$x_\phi = x\cos(\phi) - y\sin(\phi)$$

$$y_\phi = \sin(\phi)\underbrace{\cos(\theta)}_{x} + \cos(\phi)\underbrace{\sin(\theta)}_{y}$$

$$y_\phi = x\sin(\phi) + y\cos(\phi)$$

Point rotation



* Assuming a, b, x, and y are known/defined

$$x = \cos(\theta) + a$$
$$x_\phi = \cos(\phi + \theta) + a$$
$$y = \sin(\theta) + b$$
$$y_\phi = \sin(\phi + \theta) + b$$

$$\left.\begin{aligned} x_\phi &= x\cos(\phi) - y\sin(\phi) + a \\ y_\phi &= x\sin(\phi) + y\cos(\phi) + b \end{aligned}\right\} \text{rotation by point } (a, b)$$

Point rotation rule

$$x' = x\cos(\theta) - y\sin(\theta) + a$$
$$y' = x\sin(\theta) + y\cos(\theta) + b$$

In most functions replacing x and y by x' and y' (the equation equivalents) allows you to rotate the function counter-clockwise by θ